

PRE-LAB 5: Arrays and Functions

This pre-lab helps you practice arrays, functions, and basic manipulation in C. Each question is inspired by real-life scenarios and includes a ready solution.

Question 1: Sum and Average of Weekly Steps

Description: A health app tracks the number of steps a user walks each day for 7 days. Write a function that calculates the total and average steps.

Function Template:

```
int sum_steps(int steps[], int size);  
float average_steps(int steps[], int size);
```

Example:

```
Input: {4000, 5000, 3500, 6000, 4500, 7000, 3000}  
Output:  
Total steps = 33,000  
Average steps = 4714.29
```

Solution:

```
#include <stdio.h>  
  
int sum_steps(int steps[], int size) {  
    int sum = 0;  
    for(int i = 0; i < size; i++) {  
        sum += steps[i];  
    }  
    return sum;  
}  
  
float average_steps(int steps[], int size) {  
    int total = sum_steps(steps, size);  
    return (float)total / size;  
}  
  
int main() {  
    int steps[7] = {4000, 5000, 3500, 6000, 4500, 7000, 3000};  
    int total = sum_steps(steps, 7);  
    float avg = average_steps(steps, 7);  
    printf("Total steps: %d\n", total);  
    printf("Average steps: %.2f\n", avg);  
}
```

```
    return 0;
}
```

Question 2: Reverse an Array

Description: A weather app stores daily temperatures for a week. Write a function that reverses the array in place and prints it.

Function Template:

```
void reverse_array(int arr[], int size);
```

Example:

```
Input: {18, 20, 22, 21, 19, 23, 24}
Output:
```c
Original: 18 20 22 21 19 23 24
Reversed: 24 23 19 21 22 20 18
```

### Solution:

```
#include <stdio.h>

void reverse_array(int arr[], int size) {
 for(int i = 0; i < size / 2; i++) {
 int temp = arr[i];
 arr[i] = arr[size - 1 - i];
 arr[size - 1 - i] = temp;
 }
}

int main() {
 int temps[7] = {18, 20, 22, 21, 19, 23, 24};
 printf("Original: ");
 for(int i = 0; i < 7; i++) printf("%d ", temps[i]);
 printf("\n");

 reverse_array(temps, 7);

 printf("Reversed: ");
 for(int i = 0; i < 7; i++) printf("%d ", temps[i]);
 printf("\n");

 return 0;
}
```

### Question 3: Count Odd Numbers in an Array

**Description:** A sensor records integer readings. Write a function that counts the number of odd numbers in an array.

**Function Template:**

```
int count_odd(int arr[], int size);
```

**Example:**

```
Input: {3, 6, 7, 8, 9}
Output: 3
```

**Solution:**

```
#include <stdio.h>

int count_odd(int arr[], int size) {
 int count = 0;
 for(int i = 0; i < size; i++) {
 if(arr[i] % 2 != 0) count++;
 }
 return count;
}

int main() {
 int readings[5] = {3, 6, 7, 8, 9};
 printf("Number of odd readings: %d\n", count_odd(readings, 5));
 return 0;
}
```

### Question 4: Shift Array Left by One

**Description:** A delivery app stores daily package counts in an array. Write a function that shifts all elements one position to the left; the first element moves to the end.

**Function Template:**

```
void shift_left(int arr[], int size);
```

**Example:**

Input: {10, 20, 30, 40, 50}  
Output: {20, 30, 40, 50, 10}

**Solution:**

```
#include <stdio.h>

void shift_left(int arr[], int size) {
 int first = arr[0];
 for(int i = 0; i < size - 1; i++) {
 arr[i] = arr[i + 1];
 }
 arr[size - 1] = first;
}

int main() {
 int packages[5] = {10, 20, 30, 40, 50};
 shift_left(packages, 5);
 printf("Shifted array: ");
 for(int i = 0; i < 5; i++) printf("%d ", packages[i]);
 printf("\n");
 return 0;
}
```